

PROJECT BUTTON HR–V0 ELECTRICAL V3 CONNECTED CANDIDATE

Separate RESET and ARM, two PNOZ stages, dual watchdog contacts, external adapters, redundant actuator interruption.

01 External listed sources and DC boundaries

File: 01_external_sources.kicad_sch

02 Dual–channel E–stop and RESET eligibility

File: 02_estop_eligibility.kicad_sch

03 Distinct ARM and watchdog eligibility

File: 03_arm_watchdog_eligibility.kicad_sch

04 Contactor coils, mirror contacts and EDM

File: 04_contactor_edm.kicad_sch

05 Redundant actuator–power interruption

File: 05_actuator_interruption.kicad_sch

06 Protected actuator branches and central DYNAMIXEL star injection

File: 06_branches_and_injection.kicad_sch

07 Independent watchdog power, controller and drivers

File: 07_watchdog_control.kicad_sch

08 Calculated dual–channel 24 V watchdog feedback

File: 08_watchdog_feedback_interface.kicad_sch

09 Compute, debug and control terminals

File: 09_compute_and_control_terminals.kicad_sch

10 U2D2, actuator ports and bonding boundary

File: 10_actuator_interfaces.kicad_sch

11 Watchdog PCB external connectors

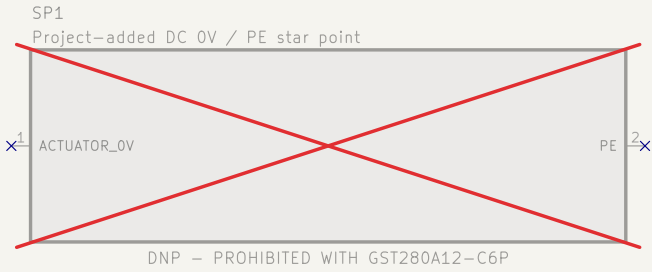
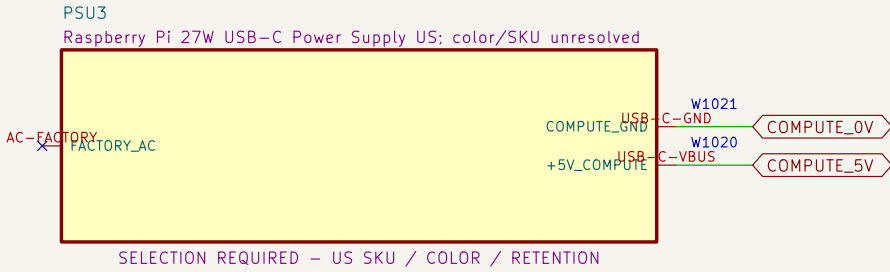
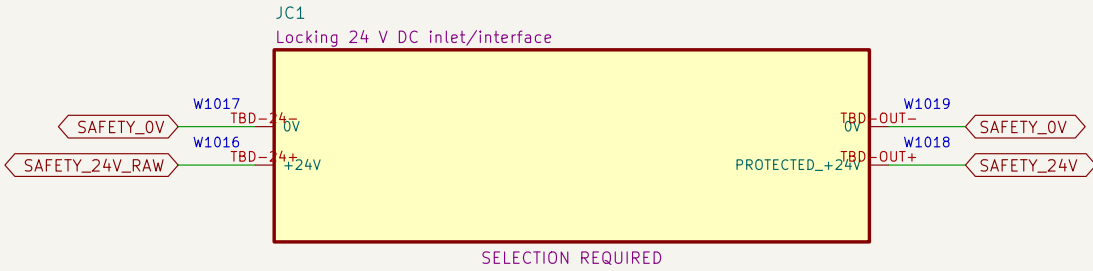
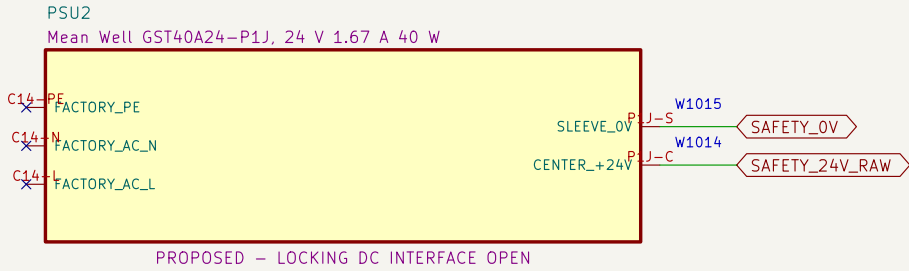
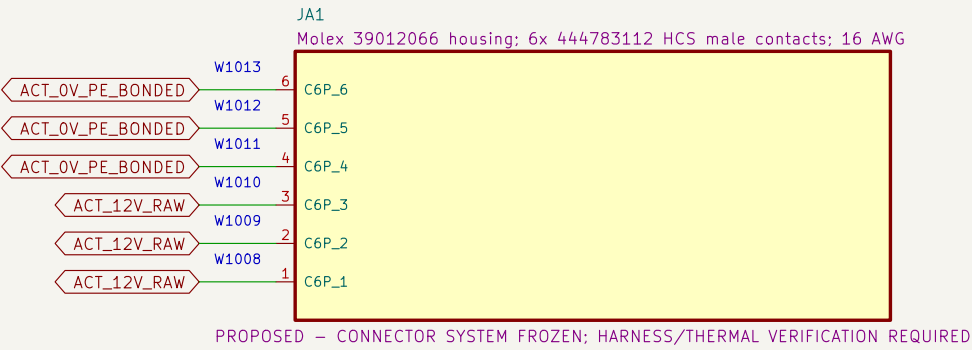
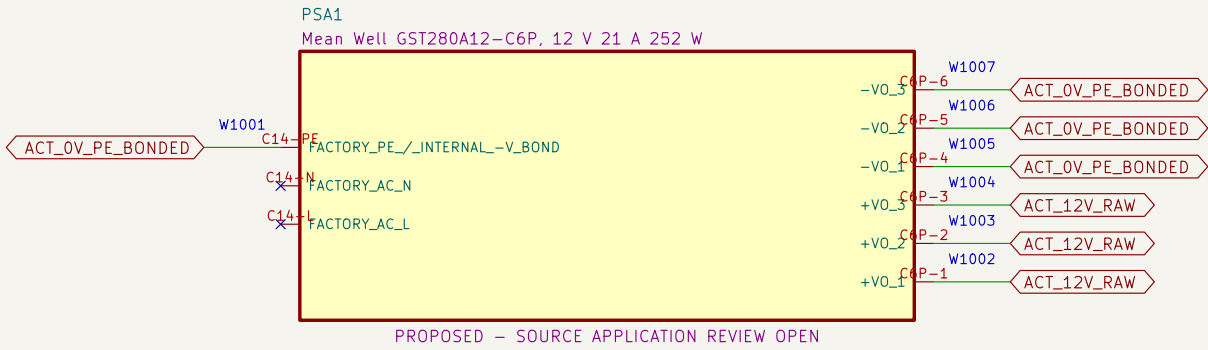
File: 11_watchdog_pcb_connectors.kicad_sch

12 Watchdog PCB test access

File: 12_watchdog_pcb_test_access.kicad_sch

01 External listed sources and DC boundaries

Factory-sealed adapters eliminate project-built mains wiring.



NOTE 1: All AC conductors shown are factory boundaries, not project-built wiring.

NOTE 2: Site cords, receptacles, GFCI/code basis and source application review remain open.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED

PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

Sheet: /01 External listed sources and DC boundaries/
File: 01_external_sources.kicad_sch

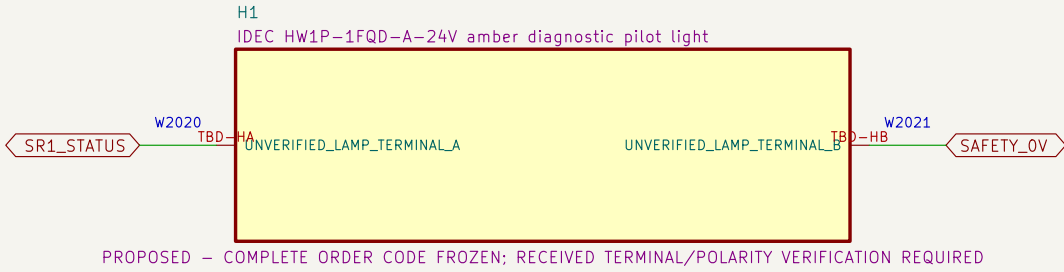
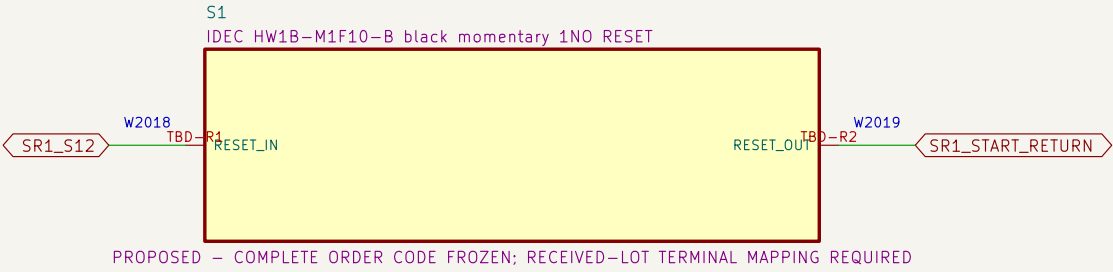
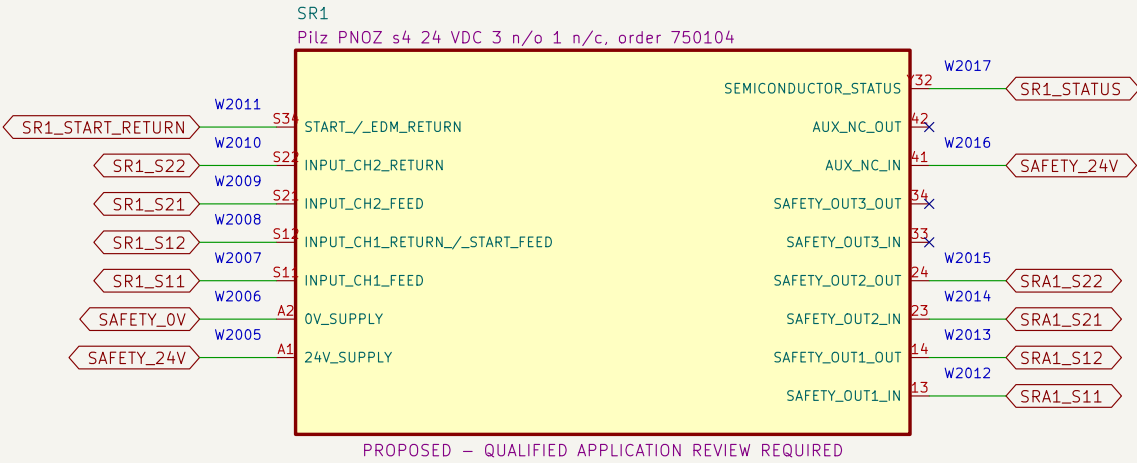
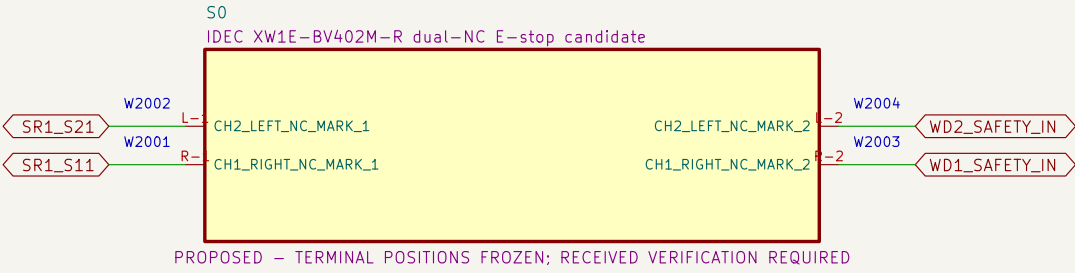
Title: PB HR-V0 V3-P1.9 – 01

Size: A3 Date: 2026-08-08 Rev: V3-P1.9

KiCad E.D.A. 10.0.5 Id: 2/13

02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains one E-stop NC and one watchdog NO contact; RESET cannot energize K1/K2.



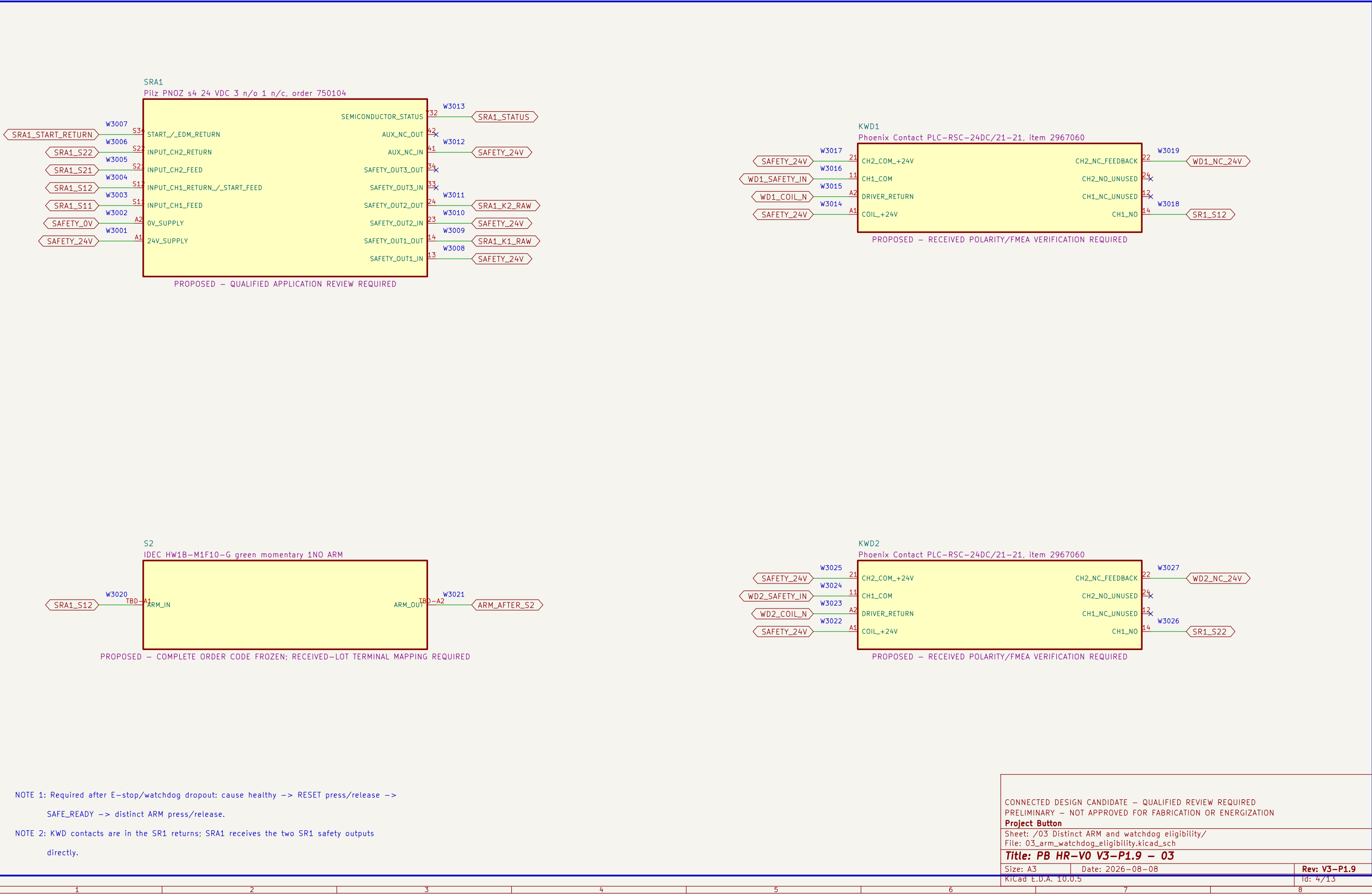
NOTE 1: Heartbeat loss opens both SR1 input returns; recovery alone cannot restore the monitored RESET stage.

NOTE 2: RESET release may make SR1 eligible, but SRA1 and K1/K2 remain de-energized until a later ARM.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /02 Dual-channel E-stop and RESET eligibility/ File: 02_estop_eligibility.kicad_sch		
Title: PB HR-V0 V3-P1.9 – 02		
Size: A3	Date: 2026-08-08	Rev: V3-P1.9
KiCad E.D.A. 10.0.5		Id: 3/13

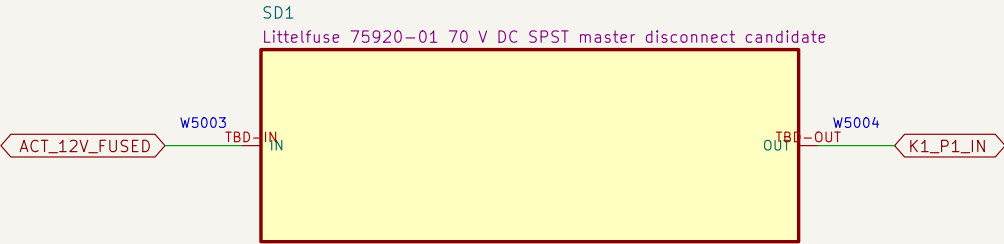
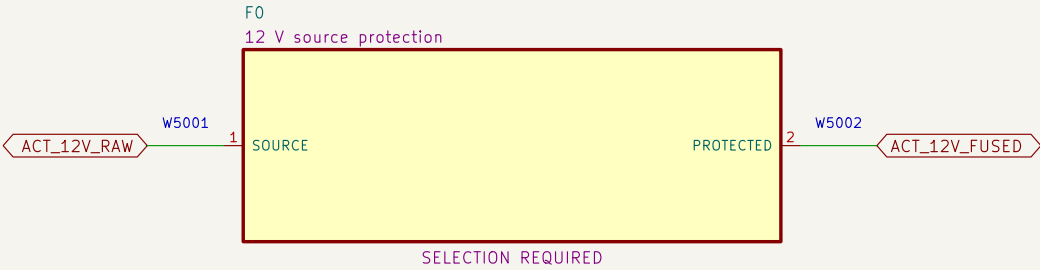
03 Distinct ARM and watchdog eligibility

SRA1 requires SR1 eligibility, two watchdog channels, EDM proof and a new ARM action.

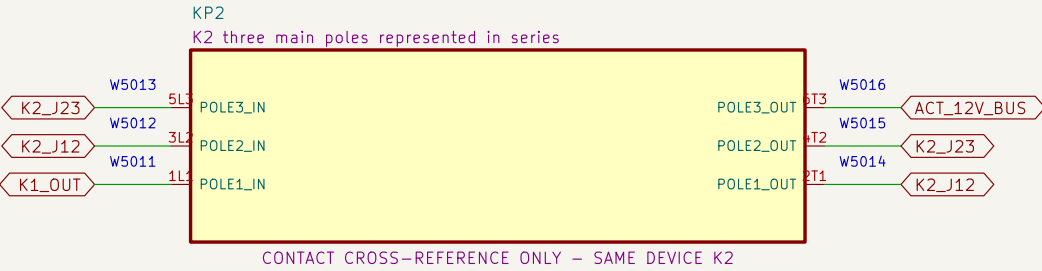
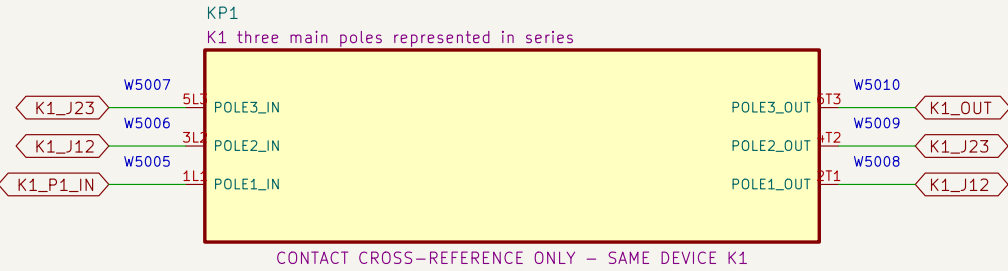


05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



PROPOSED – ORDER CODE AND SPST HIGH–SIDE TOPOLOGY FROZEN; CONDUCTOR/LUG/FAULT/LOAD–BREAK/PLACEMENT/LOCKOUT APPLICATION SELECTION REQUIRED



NOTE 1: Series jumpers: K1 2T1–>3L2, 4T2–>5L3, 6T3–>K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

Sheet: /05 Redundant actuator–power interruption/
File: 05_actuator_interruption.kicad_sch

Title: PB HR–V0 V3–P1.9 – 05

Size: A3 Date: 2026–08–08

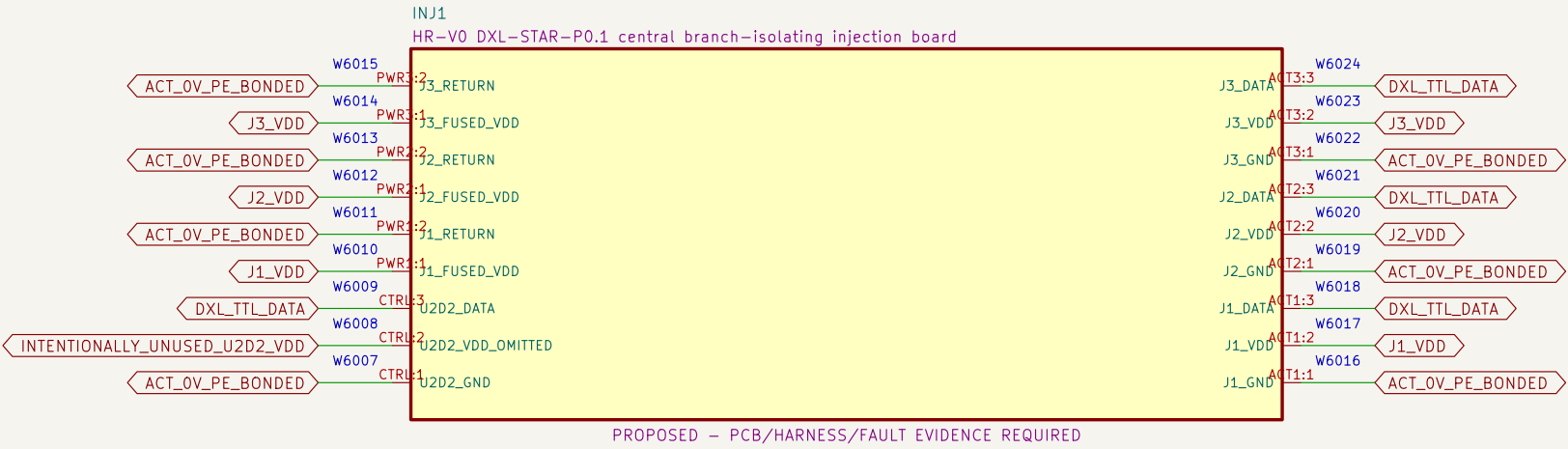
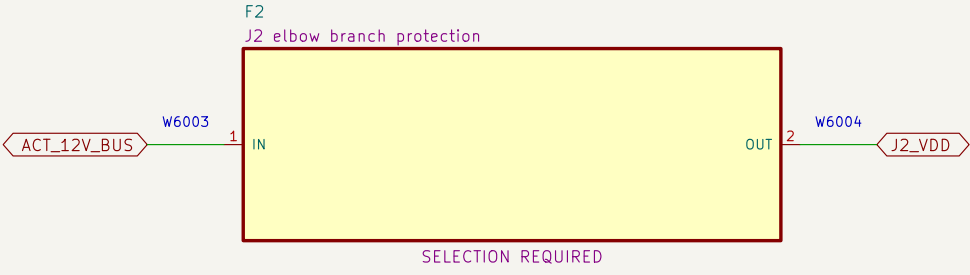
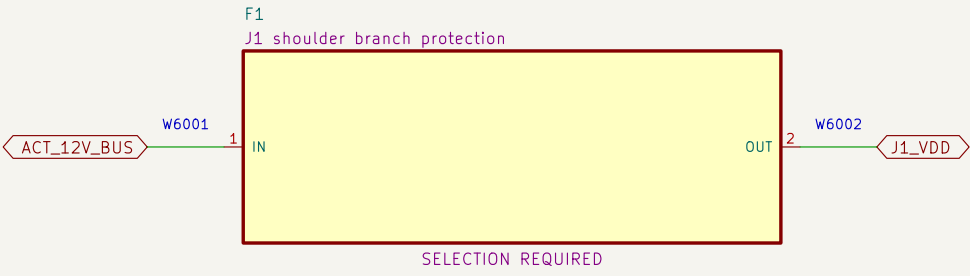
KiCad E.D.A. 10.0.5

Rev: V3–P1.9

Id: 6/13

06 Protected actuator branches and central DYNAMIXEL star injection

Each actuator has a separate protected VDD branch; U2D2 pin 2 and inter-actuator VDD paths are omitted.



NOTE 1: INJ1 is a fixed central star; U2D2 pin 2 is unrouted and J1/J2/J3 positive rails never join.

NOTE 2: Exact actuator cables, branch returns, enclosure, continuity, isolation, signal-integrity, pull and no-backfeed tests remain release gates.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

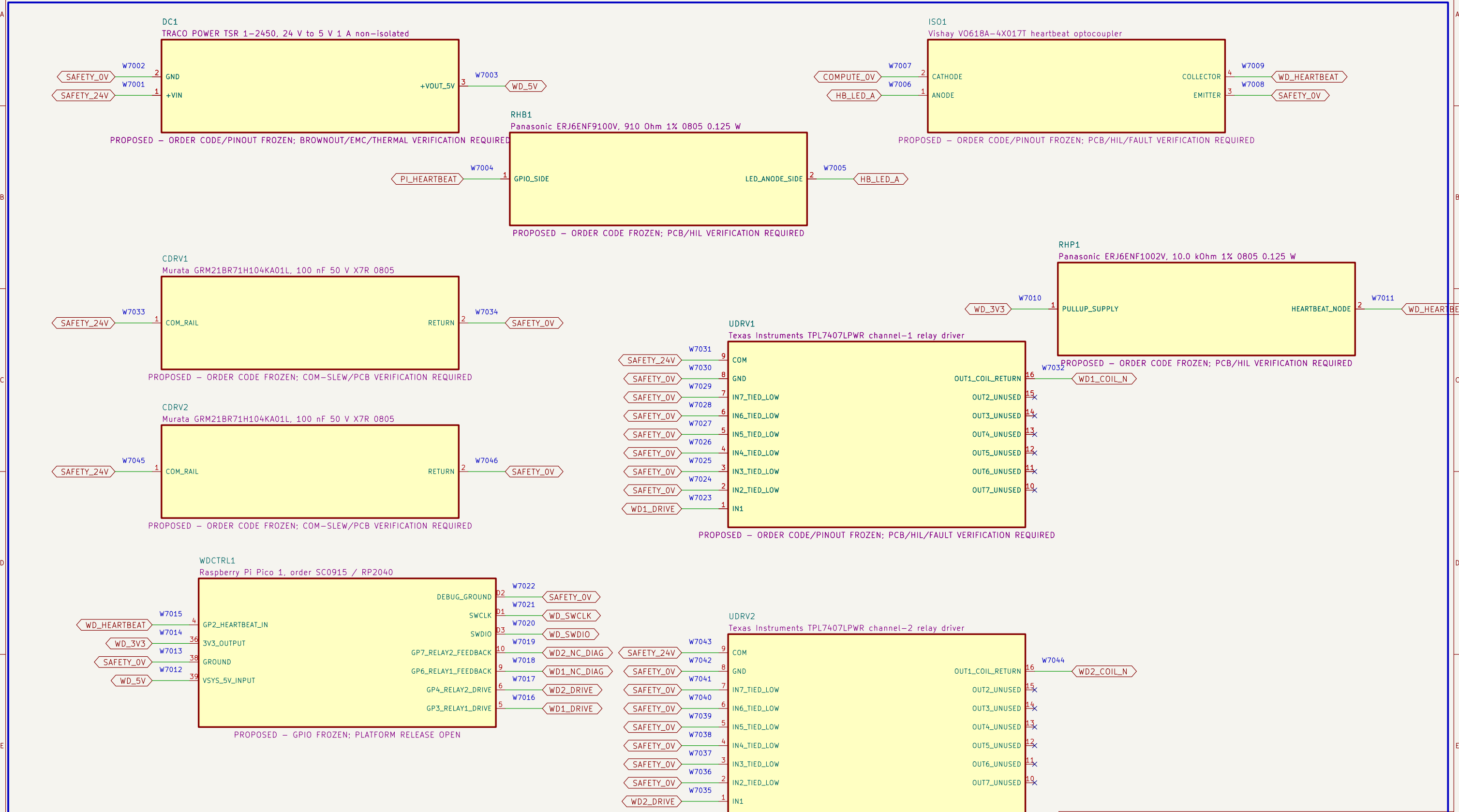
Project Button
Sheet: /06 Protected actuator branches and central DYNAMIXEL star injection/
File: 06_branches_and_injection.kicad_sch

Title: PB HR-V0 V3-P1.9 – 06

Size: A3	Date: 2026-08-08	Rev: V3-P1.9
KiCad E.D.A. 10.0.5		Id: 7/13

07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.



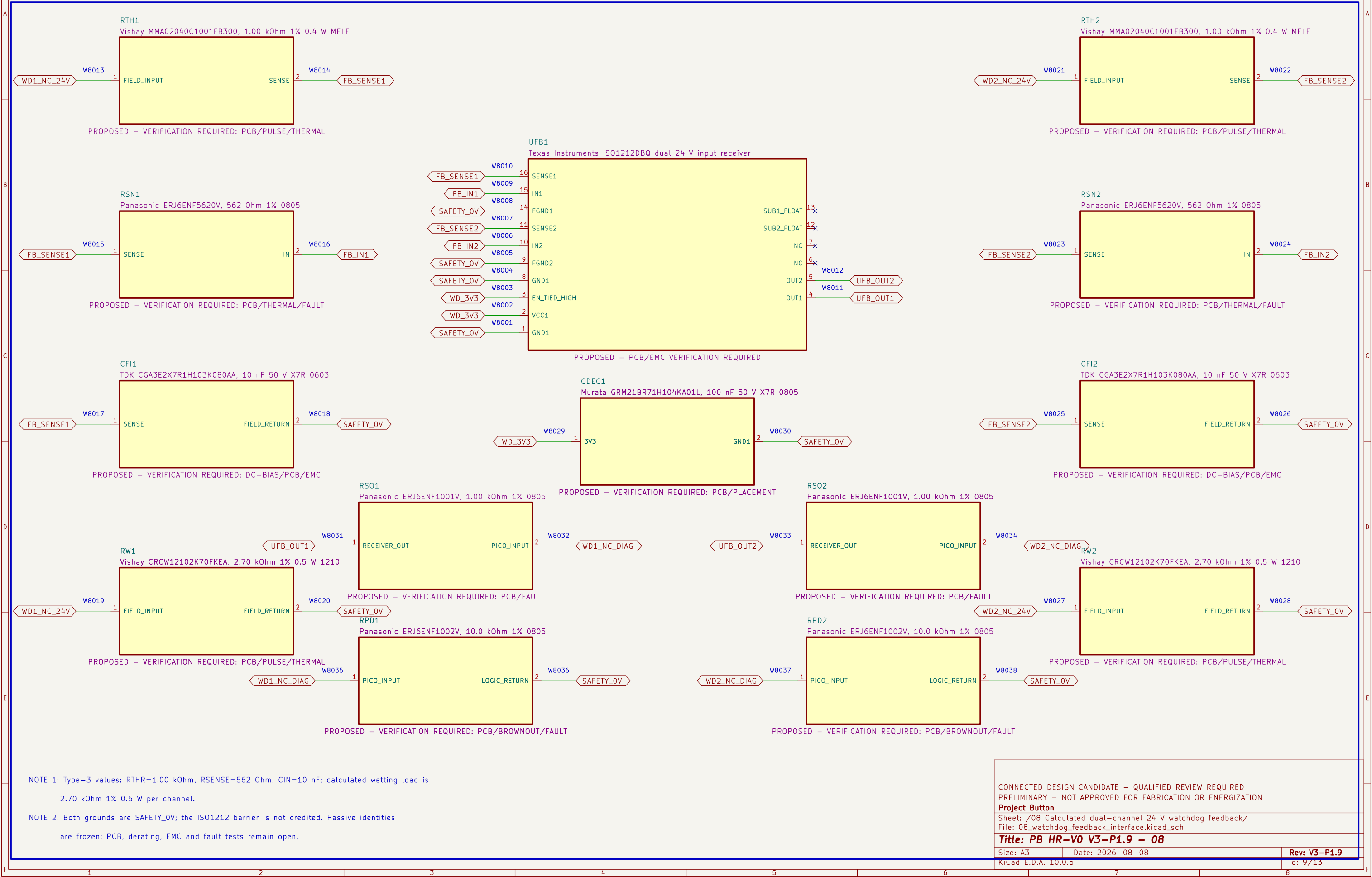
NOTE 1: Power–up, brownout, clock, stuck–GPIO and firmware tests are mandatory.

NOTE 2: Qualified review decides whether watchdog loss is credited or diagnostic only:
current safety credit is NONE.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /07 Independent watchdog power, controller and drivers/
File: 07_watchdog_control.kicad_sch
Title: PB HR–V0 V3–P1.9 – 07
Size: A3 Date: 2026–08–08 **Rev: V3–P1.9**
KiCad E.D.A. 10.0.5 Id: 8/13

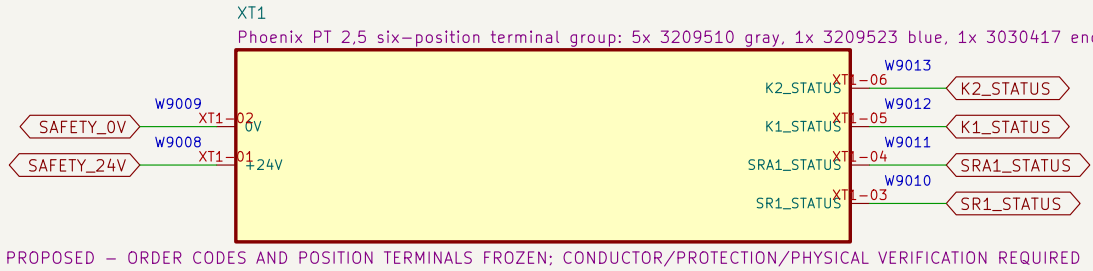
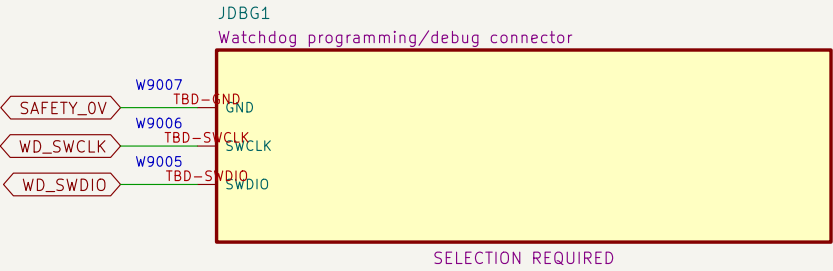
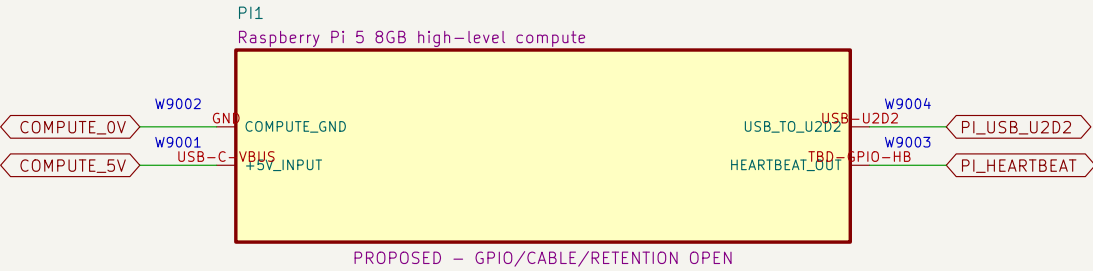
08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



09 Compute, debug and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



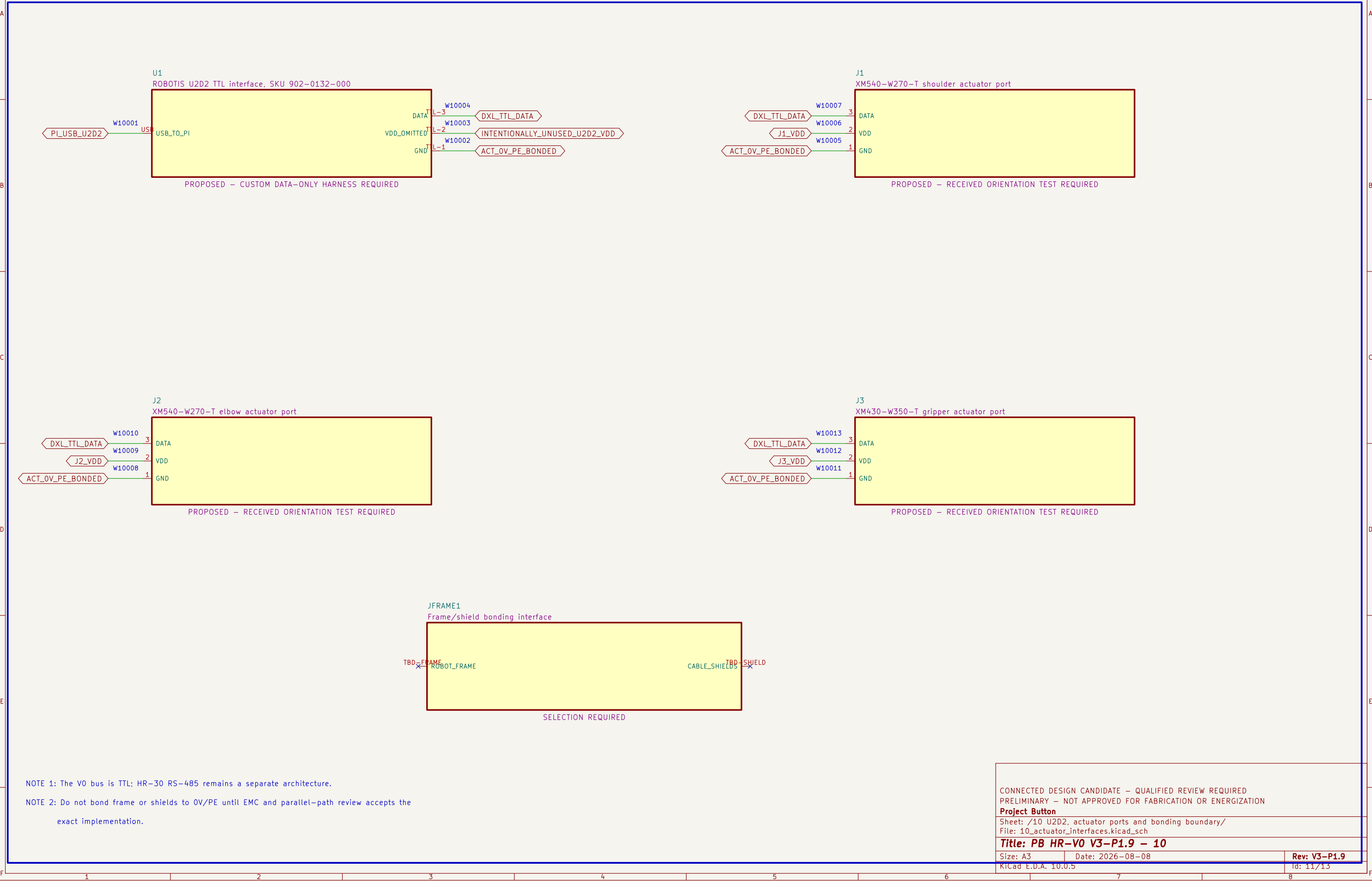
NOTE 1: Programming/debug connections must not enable outputs during operation.

NOTE 2: Terminal family, markers, conductor range, torque and enclosure layout remain selection required.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /09 Compute, debug and control terminals/ File: 09_compute_and_control_terminals.kicad_sch		
Title: PB HR-V0 V3-P1.9 – 09		
Size: A3	Date: 2026-08-08	Rev: V3-P1.9
KiCad E.D.A. 10.0.5	Id: 10/13	

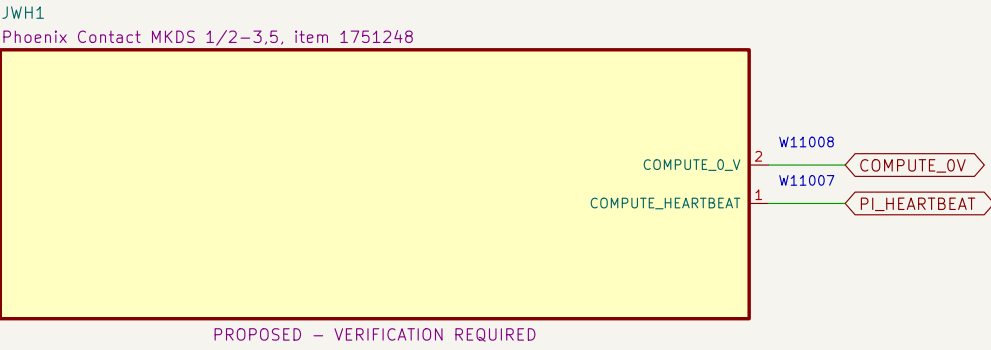
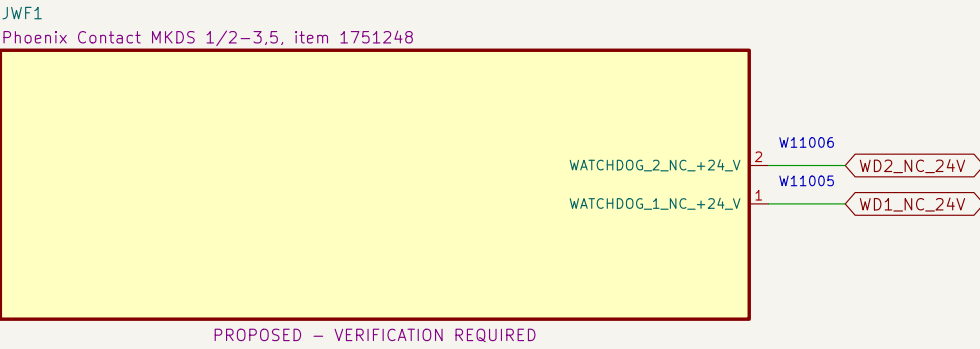
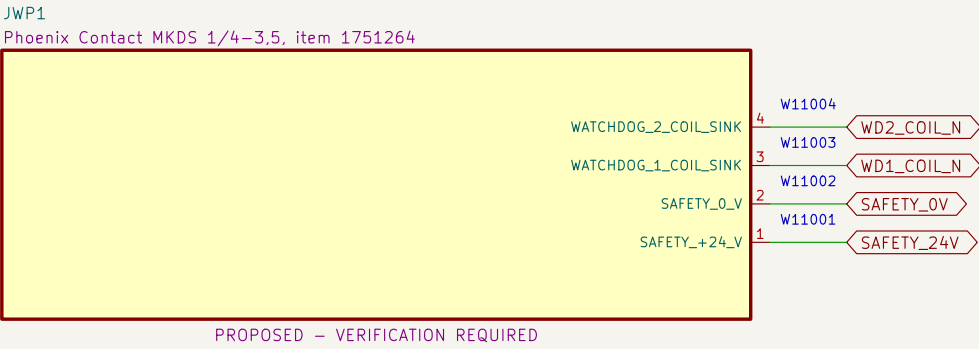
10 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.



11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.

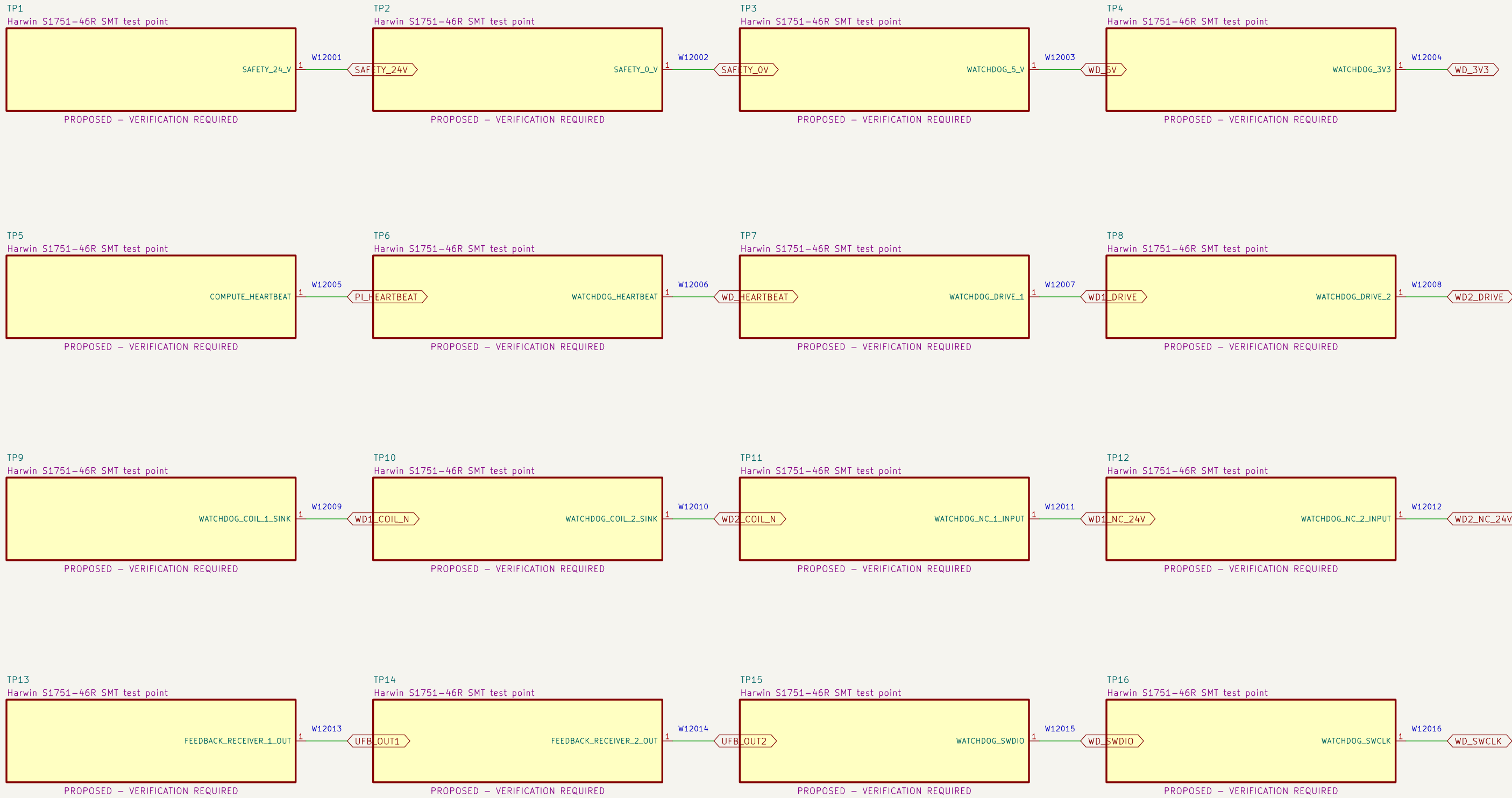


NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

12 Watchdog PCB test access

Exact clip-compatible test terminals expose controlled nodes without granting safety or operating authority.



NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

Sheet: /12 Watchdog PCB test access/
File: 12_watchdog_pcb_test_access.kicad_sch

Title: PB HR-V0 V3-P1.9 – 12

Size: A3

Date: 2026-08-08

Rev: V3-P1.9

KiCad E.D.A. 10.0.5

Id: 13/13